

ORF 524 – Final Exam

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Notes:

- Duration of examination: *7 hours*.
- All parts are equally weighted (5 points).
- Please start each question on a separate page.
- Please provide as much detail as possible in your answers.
- Answers without proper justification will not receive (partial) credit.

1 Question 1: Probability Theory and Asymptotics (15 points)

1. Let $\{X_i : 1 \leq i \leq n\}$ be i.i.d. random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $M > 0$ be a finite constant. Show that

$$\mathbb{P}[|X_i| \leq M] = 1 \quad \Longleftrightarrow \quad \mathbb{P}\left[\frac{1}{n} \sum_{i=1}^n |X_i| \leq M\right] = 1.$$

Note: The i.i.d. assumption is crucial and the result is not true in general.

- **Solution:** The “only if direction” is trivial. For the “if direction” we proceed by contradiction: suppose $\mathbb{P}[|X_i| > M] \geq p \in (0, 1)$ (a counter positive statement to $|X_i| \leq M$ (a.s.) or, equivalently, $\mathbb{P}[|X_i| \leq M] = 1$). Then,

$$\mathbb{P}\left[\frac{1}{n} \sum_{i=1}^n |X_i| > M\right] \geq \mathbb{P}\left[\min_{1 \leq i \leq n} |X_i| > M\right] = c^n,$$

which gives the contradiction

$$\mathbb{P}\left[\frac{1}{n} \sum_{i=1}^n |X_i| \leq M\right] = 1 - \mathbb{P}\left[\frac{1}{n} \sum_{i=1}^n |X_i| > M\right] < 1.$$

If the i.i.d. assumption is dropped, then the “if direction” is clearly not true in general.

2. Let $\{X_i : 1 \leq i \leq n\}$ be a random sample from $X \sim \mathbb{P}_\theta = \text{Uniform}[0, \theta]$.

Recall that $\bar{\theta}_n = 2\bar{X}_n$ is an unbiased estimator of θ . Show that

$$\hat{\theta}_n = \mathbb{E}[\bar{\theta}_n | X_{(n)}] = \frac{n+1}{n} X_{(n)}.$$

Discuss this result in connection to the Rao-Blackwell Theorem and UMVU estimation.

Hint: $\mathbb{E}[X_i | X_{(n)}, X_{(n)} \neq X_i] = X_{(n)}/2$ and $\mathbb{P}[X_{(n)} = X_i | X_{(n)}] = 1/n$.

- **Solution:** We have $\mathbb{E}[\bar{\theta}_n | X_{(n)}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i | X_{(n)}]$, where

$$\begin{aligned} \mathbb{E}[X_i | X_{(n)}] &= \mathbb{E}[X_i | X_{(n)}, X_{(n)} = X_i] \cdot \mathbb{P}[X_{(n)} = X_i | X_{(n)}] \\ &\quad + \mathbb{E}[X_i | X_{(n)}, X_{(n)} \neq X_i] \cdot \mathbb{P}[X_{(n)} \neq X_i | X_{(n)}] \end{aligned}$$

$$= X_{(n)} \cdot \frac{1}{n} + \frac{X_{(n)}}{2} \cdot \left(1 - \frac{1}{n}\right),$$

and the result follows. We are just verifying the Rao-Blackwell Theorem by direct calculation, which in this case gives the UMVU estimator because we started with a suboptimal but unbiased estimator $\bar{\theta}_n$, and then computed its conditional expectation given the complete sufficient statistic $X_{(n)}$, thereby giving the UMVU estimator $\hat{\theta}_n$ of θ .

3. Let $f(x) : \mathbb{R} \mapsto \mathbb{R}$ be a known function, and $a, b \in \mathbb{R}$ with $a < b$. Numerical integration is commonly used to approximate integrals of the form $I := \int_a^b f(x)dx$ when they are difficult (or impossible) to compute analytically. The basic procedure is as follows:

Step 1: Construct S i.i.d. uniform random variables on $[a, b]$, denoted by $\{U_i : 1 \leq i \leq S\}$.

Step 2: Approximate I by $I_S := \frac{1}{S} \sum_{i=1}^S f(U_i)(b-a)$.

Give sufficient conditions so that $I_S \rightarrow_{\mathbb{P}} I$ as $S \rightarrow \infty$.

Give an estimate of the uncertainty in the approximation.

- **Solution:** define $X_i := f(U_i)(b-a)$ for all $i = 1, 2, \dots, S$. Then, by the LLN, we get that:

$$\bar{X}_S := \frac{1}{S} \sum_{i=1}^S X_i \rightarrow_{\mathbb{P}} \mathbb{E}[X_i]$$

as long as $\mathbb{E}[X_i] < \infty$. Since $U_i \sim \text{Uniform}[a, b]$, $\mathbb{E}[X_i] = (b-a) \int_a^b f(x) \cdot \frac{1}{b-a} dx = I < \infty$.

For the second part, using the CLT and provided that $V := \mathbb{E}[X_i^2] < \infty$,

$$\frac{1}{\sqrt{S}} \sum_{i=1}^S (X_i - I) \rightarrow_d \mathcal{N}(0, V).$$

We can estimate V by the consistent and unbiased sample variance estimator,

$$\hat{V}_S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_S)^2,$$

under the usual regularity conditions.

2 Question 2: Optimal Point Estimation and Inference (50 points)

Suppose that $\{X_i : 1 \leq i \leq n\}$ is a random sample from $X \sim \mathbb{P}_\theta$, with $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$ and Lebesgue density

$$f_X(x; \theta) = a(\theta) \exp\left(-\frac{1}{2}x^2\right) \exp(b(\theta)x),$$

where $\theta \in \Theta = [0, \infty)$ is the unknown parameter, $a(\cdot)$ and $b(\cdot)$ are some functions.

1. Show that $a(\theta) = \frac{1}{\sqrt{2\pi}} \exp(-\theta/2)$ and $b(\theta) = \sqrt{\theta}$.

• **Solution:** Notice that $f_X(x; \theta)$ can be rewritten as

$$f_X(x; \theta) = \sqrt{2\pi}a(\theta) \exp\left(\frac{b(\theta)^2}{2}\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - b(\theta))^2}{2}\right).$$

Using the properties of the normal distribution,

$$\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}(x - b(\theta))^2\right) dx = \sqrt{2\pi}.$$

As a consequence, $f_X(x; \theta)$ is a density if and only if

$$a(\theta) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}b(\theta)^2\right)$$

for any $b(\theta)$. For this choice of $a(\theta)$, and for any $b(\theta)$, $f_X(x; \theta)$ is the Lebesgue density of $\mathcal{N}(b(\theta), 1)$. Thus, $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$ implies that $b(\theta) = \sqrt{\theta}$, and hence $a(\theta) = \frac{1}{\sqrt{2\pi}} \exp(-\theta/2)$.

2. Consider $\bar{X}_n^2$ as an estimator of θ , where $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Give conditions on $\{X_i : 1 \leq i \leq n\}$ under which a method of moments estimator of θ exists and equals $\hat{\theta}_{\text{MM}} = \bar{X}_n^2$.

Is $\hat{\theta}_{\text{MM}}$ unbiased?

• **Solution:** Because $X_i \sim \mathcal{N}(\sqrt{\theta}, 1)$, $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$ and $\mathbb{V}_\theta[X_i] = 1$, the estimating equation $\sqrt{\hat{\theta}_{\text{MM}}} = \bar{X}_n$ admits a solution $\hat{\theta}_{\text{MM}} = \bar{X}_n^2$ if and only if $\bar{X}_n \geq 0$.

The estimator $\bar{X}_n^2$ is biased because

$$\mathbb{E}_\theta[\bar{X}_n^2] = \left(\mathbb{E}_\theta[\bar{X}_n]\right)^2 + \mathbb{V}_\theta[\bar{X}_n] = (\mathbb{E}_\theta[X_i])^2 + \frac{1}{n} \mathbb{V}_\theta[X_i] = \theta + \frac{1}{n} \neq \theta.$$

3. Find the log likelihood function. Does θ admit a scalar sufficient statistic?

• **Solution:** Because

$$\begin{aligned}\log f_X(X_i; \theta) &= \log \left(\frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2}\theta \right) \exp \left(-\frac{1}{2}X_i^2 \right) \exp \left(\sqrt{\theta}X_i \right) \right) \\ &= -\frac{1}{2} \log(2\pi) - \frac{1}{2}\theta - \frac{1}{2}X_i^2 + \sqrt{\theta}X_i,\end{aligned}$$

the log likelihood function is

$$\ell_n(\theta) = \sum_{i=1}^n \log f_X(X_i; \theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2}\theta - \frac{1}{2} \sum_{i=1}^n X_i^2 + \sqrt{\theta} \sum_{i=1}^n X_i.$$

It follows from the factorization theorem that the scalar $n\bar{X}_n = \sum_{i=1}^n X_i$ is a sufficient statistics for θ .

4. Show that the maximum likelihood estimator of θ is $\hat{\theta}_{\text{ML}} = \left(\max\{\bar{X}_n, 0\} \right)^2$.

• **Solution:** If $\bar{X}_n \leq 0$, then $\ell_n(\theta)$ is decreasing on Θ and thus $\hat{\theta}_{\text{ML}} = \min\{\theta : \theta \in \Theta\} = 0$.

On the other hand, $\bar{X}_n > 0$, then $\hat{\theta}_{\text{ML}} = \bar{X}_n^2$ because

$$\ell_{\theta,n}(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = -\frac{n}{2} + \frac{1}{2\sqrt{\theta}} \sum_{i=1}^n X_i = 0 \quad \Longleftrightarrow \quad \theta = \bar{X}_n^2$$

and

$$\ell_{\theta,n}(\theta) = \frac{\partial^2}{\partial \theta^2} \ell_n(\theta) = -\frac{1}{4\theta^{3/2}} \sum_{i=1}^n X_i < 0.$$

As a consequence, $\hat{\theta}_{\text{ML}} = \left(\max\{\bar{X}_n, 0\} \right)^2$.

5. Find the uniform minimum variance unbiased estimator $\hat{\theta}_{\text{UMVU}}$ of θ .

• **Solution:** The sufficient statistic $n\bar{X}_n = \sum_{i=1}^n X_i$ is complete because

$$\{b(\theta) : \theta \in \Theta\} = \{\sqrt{\theta} : \theta \geq 0\} = [0, \infty)$$

contains an open set. It was shown above that $\mathbb{E}_\theta[\bar{X}_n^2] = \theta + 1/n$. Therefore,

$$\hat{\theta}_{\text{UMVU}} = \bar{X}_n^2 - \frac{1}{n}$$

is an unbiased estimator based on the complete, sufficient statistic $n\bar{X}_n = \sum_{i=1}^n X_i$.

6. Show that the Cramér-Rao bound for unbiased estimators of θ is $\text{CRLB}(\theta) = 4\theta/n$.

Is $\text{CRLB}(\theta)$ attained by the estimator $\hat{\theta}_{\text{UMVU}}$?

Hint: If $Y \sim \text{Normal}(\mu, \sigma^2)$, then $\mathbb{V}[Y^2] = 4\mu^2\sigma^2 + 2\sigma^4$.

• **Solution:** The Cramér-Rao bound for unbiased estimators is

$$\text{CRLB}(\theta) = \frac{1}{n\mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f_X(X_i; \theta) \right)^2 \right]}.$$

Because

$$\frac{\partial}{\partial \theta} \log f_X(X_i; \theta) = -\frac{1}{2} + \frac{1}{2\sqrt{\theta}} X_i$$

and $X_i \sim \text{N}(\sqrt{\theta}, 1)$, we have

$$\frac{\partial}{\partial \theta} \log f_X(X_i; \theta) \sim \text{N}\left(0, \frac{1}{4\theta}\right),$$

which gives $\text{CRLB}(\theta) = 4\theta/n$.

Finally, because $\bar{X}_n \sim \text{N}(\sqrt{\theta}, 1/n)$, it follows from the hint that

$$\mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}] = \mathbb{V}_\theta[\bar{X}_n^2] = \frac{4\theta}{n} + \frac{2}{n^2},$$

which implies that $\hat{\theta}_{\text{UMVU}}$ does not attain the bound $\text{CRLB}(\theta)$.

7. Let $\theta_0 > 0$ be some constant and consider the one-sided testing problem:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0.$$

(a) Consider a test which rejects H_0 if and only if $\bar{X}_n > c$, where c is some constant (possibly depending on θ_0). Find c such that the testing procedure has 5% size.

- **Solution:** We have

$$\begin{aligned}
\mathbb{P}_{\theta_0}[\bar{X}_n > c] &= 1 - \mathbb{P}_{\theta_0}[\bar{X}_n \leq c] \\
&= 1 - \mathbb{P}_{\theta_0}[\sqrt{n}(\bar{X}_n - \sqrt{\theta_0}) \leq \sqrt{n}(c - \sqrt{\theta_0})] \\
&= 1 - \Phi\left(\sqrt{n}(c - \sqrt{\theta_0})\right),
\end{aligned}$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution. As a consequence, a test has size 5% if and only if

$$\sqrt{n}(c - \sqrt{\theta_0}) = \Phi^{-1}(0.95) \quad \Longleftrightarrow \quad c = \sqrt{\theta_0} + \Phi^{-1}(0.95)/\sqrt{n}.$$

- (b) Show that the testing procedure derived above is uniformly most powerful (within the class of tests of the same level).

- **Solution:** For any $\theta_1 > \theta_0$, the log likelihood ratio is

$$\ell_n(\theta_1) - \ell_n(\theta_0) = -\frac{n}{2}(\theta_1 - \theta_0) + n(\sqrt{\theta_1} - \sqrt{\theta_0})\bar{X}_n,$$

which is an increasing function of $\bar{X}_n$. Therefore, it follows from the Neyman-Pearson lemma that the test derived above is the (uniformly) most powerful 5% test of $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$. By the Karlin-Rubin lemma, it follows that the test is also the uniformly most powerful 5% test of $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$.

- (c) Derive the power envelop of the 5% level hypothesis test above.

- **Solution:** The power function is

$$\begin{aligned}
\beta(\theta) &= \mathbb{P}_{\theta}[\bar{X}_n > \sqrt{\theta_0} + \Phi^{-1}(0.95)/\sqrt{n}] \\
&= 1 - \mathbb{P}_{\theta}[\sqrt{n}(\bar{X}_n - \sqrt{\theta}) \leq \Phi^{-1}(0.95) + \sqrt{n}(\sqrt{\theta_0} - \sqrt{\theta})] \\
&= 1 - \Phi\left(\Phi^{-1}(0.95) + \sqrt{n}(\sqrt{\theta_0} - \sqrt{\theta})\right).
\end{aligned}$$

- (d) Construct a 95% confidence interval for θ based on the UMP test derived above.

- **Solution:** Using test inversion, we have $\{\theta \in \Theta : \bar{X}_n \leq \sqrt{\theta} + \Phi^{-1}(0.95)/\sqrt{n}\}$.

3 Question 3: Misspecified Maximum Likelihood (35 points)

Let $\{X_i : 1, 2, \dots, n\}$ be a random sample from $X \sim \mathbb{P}_{\theta_0}$ with $\mathbb{P}_{\theta_0} \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, where $F(x) = \mathbb{P}_\theta[X_i \leq x]$ is absolutely continuous with Lebesgue density $f(x; \theta)$. Let $f(x) = f(x; \theta_0)$ to save notation. Consider another family of Lebesgue densities, $\{g(x; \theta) : \theta \in \Theta \subseteq \mathbb{R}\}$, which may not be equal to $f(x; \theta)$. A *possibly misspecified* Maximum Likelihood Estimator (MLE) is given by

$$\check{\theta}_n = \arg \max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \log g(X_i; \theta),$$

and the associated population optimization problem is

$$\check{\theta}_0 = \arg \max_{\theta \in \Theta} \mathbb{E}_{\theta_0}[\log g(X; \theta)] = \arg \max_{\theta \in \Theta} \int [\log g(x; \theta)] f(x) dx.$$

assuming the “usual” regularity conditions hold. You will be asked to state and impose additional assumptions (or regularity conditions) as needed below.

1. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0} \left[\log \left(\frac{g(X; \theta)}{f(X)} \right) \right] = \int \log \left(\frac{g(x; \theta)}{f(x)} \right) f(x) dx \leq 0,$$

with equality if and only if $g(x; \theta) = f(x)$.

Use this result to justify the following interpretation for the pseudo-true value $\check{\theta}_0$:

$$\check{\theta}_0 = \arg \min_{\theta \in \Theta} D(g(\cdot, \theta), f), \quad D(g(\cdot, \theta), f) = \int [\log f(x)] f(x) dx - \int [\log g(x; \theta)] f(x) dx.$$

The “metric” $D(g(\cdot, \theta), f)$ is called the *Kullback-Leibler* (K-L) distance. Can you give an interpretation to the (possibly) misspecified MLE based on this criterion?

- **Solution:** Assuming we can interchange derivatives and expectations, by Jensen’s inequality and concavity of the $\log(\cdot)$ function,

$$\mathbb{E}_{\theta_0} \left[\log \left(\frac{g(X; \theta)}{f(X)} \right) \right] \leq \log \left(\int \frac{g(x, \theta)}{f(x)} f(x) dx \right) = \log \left(\int g(x, \theta) dx \right) = \log 1 = 0.$$

$D(g(\cdot, \theta), f)$ can be considered as a measure of fit between the two distributions, the closer this quantity is to zero the better the fit. The Kullback-Leibler criterion is commonly used for model selection. Note that $\check{\theta}_0 = \arg \min_{\theta \in \Theta} D(g(\cdot, \theta), f)$ implies that $g(\cdot, \check{\theta}_0)$ is the best fitting distribution using the K-L criterion. This does not mean it is the best fitting distribution according to another criterion. Indeed, if we used a different distance measure, we would likely obtain a different best fitting distribution. There are many different information criterion. The motivation for the KL criterion comes from the likelihood. However, in the model misspecification setup there are alternative methods to likelihood methods, which can be used to find the best fitting distribution (these methods may be more robust, such as the Renyi information criterion).

2. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0} \left[\frac{\partial}{\partial \theta} \log g(X; \theta) \Big|_{\theta=\check{\theta}_0} \right] = 0.$$

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

• **Solution:** since

$$\check{\theta}_0 = \arg \max_{\theta \in \Theta} \mathbb{E}_{\theta_0} [\log g(X; \theta)],$$

assuming differentiability,

$$0 = \frac{\partial}{\partial \theta} \mathbb{E}_{\theta_0} [\log g(X; \theta)]|_{\theta=\check{\theta}_0} = \mathbb{E}_{\theta_0} \left[\frac{\partial}{\partial \theta} \log g(X; \theta) \Big|_{\theta=\check{\theta}_0} \right]$$

where the second equality assumes interchangeability of derivatives and expectations.

When $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$, we get $\check{\theta}_0 = \theta_0$.

3. Providing appropriate regularity conditions, show that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) = -\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + o_{\mathbb{P}}(1) \rightarrow_d \mathcal{N}(0, V(\check{\theta}_0)),$$

where $\mathbb{E}_{\theta_0}[\psi(X_i)] = 0$ and $\mathbb{E}_{\theta_0}[\psi(X_i)^2] = V(\check{\theta}_0) = H(\check{\theta}_0)^{-1}\Sigma(\check{\theta}_0)H(\check{\theta}_0)^{-1}$.

Give the exact form of the influence function $\psi(x)$.

Give the exact form of the variance components $H(\theta)$ and $\Sigma(\theta)$.

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

- **Solution:** start by assuming that

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \log g(X_i; \theta) - \mathbb{E}_{\theta_0}[\log g(X; \theta)] \right| = o_{\mathbb{P}}(1)$$

to ensure consistency of $\check{\theta}_n$ for $\check{\theta}_0$. Then, assuming differentiability,

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_n) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) + \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \check{\theta}_n) (\check{\theta}_n - \check{\theta}_0) \end{aligned}$$

where $\check{\theta}_n \rightarrow_{\mathbb{P}} \check{\theta}_0$ by consistency of $\check{\theta}_n$. Now, define

$$H_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta), \quad H(\theta) = \mathbb{E}_{\theta_0} \left[\frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta) \right]$$

and suppose

$$\sup_{\theta \in \Theta} |H_n(\theta) - H(\theta)| = o_{\mathbb{P}}(1).$$

Then,

$$\begin{aligned} \left| H_n(\check{\theta}_n) - H(\theta_0) \right| &\leq \left| H_n(\check{\theta}_n) - H(\check{\theta}_n) \right| + \left| H(\check{\theta}_n) - H(\theta_0) \right| \\ &\leq \sup_{\theta \in \Theta} |H_n(\theta) - H(\theta)| + \left| H(\check{\theta}_n) - H(\theta_0) \right| \\ &= o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1) \end{aligned}$$

where the first term goes to zero by assumption and the second one goes to zero by

consistency of $\tilde{\theta}_n$. Therefore,

$$\begin{aligned}\sqrt{n}(\check{\theta}_n - \check{\theta}_0) &= \left[\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \tilde{\theta}_n) \right]^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) \\ &= H_0^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) + o_{\mathbb{P}}(1)\end{aligned}$$

by the CMT and using that the second sum is $O_{\mathbb{P}}(1)$ and where $H_0 = H(\check{\theta}_0)$. It follows that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + o_{\mathbb{P}}(1)$$

with

$$\psi(x) = H_0^{-1} \frac{\partial}{\partial \theta} \log g(x; \check{\theta}_0), \quad \mathbb{V}_{\theta_0}[\psi(X_i)] = H_0^{-1} \Sigma_0 H_0^{-1}, \quad \Sigma(\theta) = \mathbb{E}_{\theta_0} \left[\left(\frac{\partial}{\partial \theta} \log g(x; \theta) \right)^2 \right]$$

Finally, when $g(x; \check{\theta}_0) = f(x)$ or $g(x; \theta) = f(x; \theta)$, the sandwich variance reduces to the inverse of the information, that is, $V_0 = \mathcal{I}_0^{-1}$.

4. Propose a consistent estimator of the asymptotic variance $V(\check{\theta}_0)$.

Construct a feasible, asymptotically pivotal statistic.

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

- **Solution:** By consistency of $\check{\theta}_n$, we estimate the variance terms consistently using $\hat{H} = \hat{H}(\check{\theta}_n)$ and $\hat{\Sigma} = \hat{\Sigma}(\check{\theta}_n)$, where

$$\hat{H}(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta) \quad \text{and} \quad \hat{\Sigma}(\theta) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\partial}{\partial \theta} \log g(x; \theta) \right)^2.$$

We can use two equivalent asymptotically pivotal statistics,

$$T(\check{\theta}_0) = \hat{V}^{-1/2} \sqrt{n}(\check{\theta}_n - \check{\theta}_0) \rightarrow_d \mathcal{N}(0, 1)$$

and

$$Q(\check{\theta}_0) = n(\check{\theta}_n - \check{\theta}_0)' \hat{V}^{-1} (\check{\theta}_n - \check{\theta}_0) \rightarrow_d \chi_1^2,$$

where $\hat{V} = \hat{H}^{-1} \hat{\Sigma} \hat{H}^{-1}$.

5. Suppose that $X_i = \mathbb{1}(\theta_0 - \epsilon_i \geq 0)$ with $\{\epsilon_i : 1, 2, \dots, n\}$ be a random sample from $\epsilon \sim \mathcal{N}(0, 1)$.

(a) Show that $\mathcal{P} = \{\text{Bernoulli}(\Phi(\theta)) : \theta \in \Theta \subseteq \mathbb{R}\}$.

Find the (correctly specified) maximum likelihood estimator $\hat{\theta}_{\text{ML}}$ of θ_0 .

Find the asymptotic distribution of $\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta_0)$, and give the exact form of the asymptotic variance component(s).

• **Solution:** we have that $X_i \sim \text{Bernoulli}(p_0)$ where $p_0 = \Phi(\theta_0)$. The log-likelihood is given by

$$\ell_n(\theta) = \log \Phi(\theta) \sum_{i=1}^n X_i + \log(1 - \Phi(\theta)) \left(n - \sum_{i=1}^n X_i \right)$$

The FOC is

$$\frac{\bar{X}_n}{\Phi(\hat{\theta})} - \frac{(n - \bar{X}_n)}{1 - \Phi(\hat{\theta})} = 0 \quad \Rightarrow \quad \hat{\theta}_{\text{ML}} = \Phi^{-1}(\bar{X}_n)$$

By the delta method, given that $\mathbb{E}_{\theta_0}[X_i] = \Phi(\theta_0)$ and $\mathbb{V}_{\theta_0}[X_i] = \Phi(\theta_0)(1 - \Phi(\theta_0))$,

$$\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta_0) \rightarrow_d \mathcal{N}(0, g'(\Phi(\theta_0))^2 \Phi(\theta_0)(1 - \Phi(\theta_0)))$$

where

$$g(z) = \Phi^{-1}(z), \quad g'(z) = \frac{1}{\phi(\Phi^{-1}(z))}, \quad g'(\Phi(\theta_0)) = \frac{1}{\phi(\theta_0)}$$

so the asymptotic variance is

$$V_0 = \frac{\Phi(\theta_0)(1 - \Phi(\theta_0))}{\phi(\theta_0)^2}$$

(b) Suppose a researcher employs the misspecified model $\epsilon \sim \text{Uniform}(0, 1)$.

Find the misspecified maximum likelihood estimator $\check{\theta}_n$ of $\check{\theta}_0$, and compute the form of pseudo-true value $\check{\theta}_0$ in this case.

Find the asymptotic distribution of $\sqrt{n}(\check{\theta}_n - \check{\theta}_0)$, and give the exact form of the asymptotic variance component(s).

• **Solution:** under the misspecified model $g(x, \theta) = \theta^x(1 - \theta)^{1-x}$. The misspecified

MLE is solves

$$\frac{\bar{X}_n}{\hat{\theta}} - \frac{1 - \bar{X}_n}{1 - \hat{\theta}} = 0 \quad \Rightarrow \quad \check{\theta}_n = \bar{X}_n$$

On the other hand, $\check{\theta}_0$ solves

$$\frac{\Phi(\theta_0)}{\check{\theta}_0} - \frac{1 - \Phi(\theta_0)}{1 - \check{\theta}_0} = 0 \quad \Rightarrow \quad \check{\theta}_0 = \Phi(\theta_0)$$

It follows by the CLT that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) \rightarrow_d \mathcal{N}(0, V), \quad V = \Phi(\theta_0)(1 - \Phi(\theta_0)).$$